

Noor Khial

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SUMMARY

AI/ML engineer and researcher with experience in designing and deploying production machine learning systems for **logistics and customs operations**. Holds an MSc in Computing from Qatar University, with research on online learning algorithms for autonomous aerial navigation.

Fun fact: one reviewer left 99 comments on a single journal paper submission of mine. Since the journal was among the most cited in AI, I responded with a 60+ page rebuttal and the paper was accepted. Persistence pays off.

INDUSTRY EXPERIENCE

- **AI/ML Engineer** May 2025 – Present
Malomatia — Authority of Customs, Qatar (Al-Nadeeb System)
 - **HSSify Engine — HS Code Classification System** (*1st place, Malomatia AI Contest*). in-house AI system that predicts HS code of imported goods. Built on *MS Azure* with fine-tuned BERT transformer model. Currently in production within the Al-Nadeeb customs platform, where it achieves **92% accuracy** and an **80% efficiency improvement** over manual classification, measured on live production traffic.
 - **AI-Based Goods Valuation System** (*in progress*). end-to-end ML pipeline that predicts the price of imported items. The pipeline operates in three stages, *Data acquisition*, *Model training* using meta learning, and *Price prediction*. Current staging performance: **71% accuracy** with a **60% improvement** over the manually configured baseline.
 - **Stack:** MS Azure, Web Services, PyTorch, Hugging Face Transformers, BERT, Docker, SQL, Python.
- **AI Engineer** Feb 2025 – Apr 2025
Applab
 - Built multi-agent AI chatbots for Qatari national institutes on Microsoft Azure, integrating Azure OpenAI Service with retrieval and orchestration components.
- **Full Stack Intern** May 2022 – Jul 2022
QCRI, Cybersecurity Division (CIBR)
 - Developed a responsive web application for personalized investment portfolio generation using ReactJS and RESTful APIs.

EDUCATION

- **MSc in Computing** Jan 2023 – Dec 2024
Qatar University GPA: 3.77/4.0
 - *Thesis:* Smart Path Planning for Mobile Multi-Target Visitation in UAV Missions in Stochastic & Adversarial Environments
 - *Supervisor:* [Prof. Amr Mohamed](#), Professor of Computer Engineering and Head of the CSE Department at Qatar University.
 - *Funding:* Graduate Research Assistantship (tuition + stipend)
 - *Award:* Research Distinguished Award
 - *Coursework:* Algorithms, Deep Learning, Information Retrieval, Computer Vision
- **BSc in Computer Engineering** Jan 2018 – May 2022
Qatar University GPA: 3.5/4.0
 - *Senior Project:* Design and Implementation of an Integrated Onboard Computer & Transceiver Subsystem for [QUBESat](#), the first CubeSat launched by Qatar under Qatar University.
 - *Supervisor:* [Prof. Tamer Khattab](#), Professor of Electrical Engineering, Head of the Intelligent Information Processing Lab (IIPL), and the Excellence in Teaching Department at Qatar University.
 - *Funding:* The Amiri Diwan Scholarship (merit-based, full tuition)
 - *Coursework:* Probability & Statistics, Discrete Mathematics, Machine Learning

PUBLICATIONS

Peer-Reviewed Journal Papers

- [J1] **Khial, N.**, Allahham, M.S., Mhaisen, N., Ismail, L., Mabrok, M., and Mohamed, A., “DRONE-RL: Dynamic reinforcement learning for online navigation of UAVs in evolving environments,” *Knowledge-Based Systems*, Elsevier, vol. 334, p. 115147, 2026.
- [J2] **Khial, N.**, Mhaisen, N., Mabrok, M., and Mohamed, A., “An online learning framework for UAV search mission in adversarial environments,” *Expert Systems with Applications*, Elsevier, vol. 267, p. 126136, 2025.

Conference Papers

- [C1] **Khial, N.**, Mhaisen, N., Ismail, L., Mabrok, M., and Mohamed, A., “Multi-Target Path Planning with Probabilistic Detection in Cluttered Environments,” *IEEE International Conference on Communications (ICC)*, pp. 1298–1303, 2025.

RESEARCH EXPERIENCE

- Part-Time Researcher** *Qatar University* Sep 2025 – Present
 - Applying new online learning theory to autonomous navigation.
- Graduate Research Assistant** *Qatar University* Jan 2023 – Jan 2025
 - Contributed to multiple funded research projects.
 - Designed online learning and deep RL algorithms for UAV navigation in non-stationary settings.
 - Published 2 journal papers in top ranked AI journals.
- NLP Research Intern** *QCRI, Arabic Language Technologies* May 2021 – Jul 2021
 - Fine-tuned AraBERT for Arabic propaganda detection, achieving **85% accuracy**.
 - Built an automated data collection and preprocessing pipeline for Arabic text.
- Research Intern** *Center for Advanced Materials (CAM)* Jun 2020 – Jul 2020

OPEN-SOURCE RESEARCH SOFTWARE

- Arabic Information Retrieval System** [GitHub](#)
Semantic search system combining the FAISS engine with the mContriever deep model for Arabic document retrieval.
Python, FAISS, PyTorch, Hugging Face Transformers, OpenAI API
- UAV Navigation in Adversarial Environments** [GitHub](#)
Deep reinforcement learning implementation with a custom simulation environment for multi-target tracking and path optimization.
Python, TensorFlow, OpenAI Gym

ACADEMIC SERVICE

- Technical Program Committee Member:** IEEE WCNC, IEEE ENERGYCON, IEEE ICC
- Conference Volunteer:** ASTROCON-2021, IEEE-ICIOT-2020

TEACHING EXPERIENCE

- Academic Tutor** *Qatar University* Sep 2020 – May 2021
 - Tutored undergraduate students in Calculus II & III and Probability & Statistics.

AWARDS & HONORS

- 1st Place, Malomatia AI Contest** – HSSify Engine, 2025
- Graduate Research Assistantship** – Qatar University, 2023–2024
- The Amiri Diwan Scholarship** – 2018–2022
- Dean’s List** – Qatar University, College of Engineering, 2019–2022
- QU Undergraduate Startups Pre-Incubation Program** – Qatar University, 2022

TECHNICAL PROFICIENCIES

Research Areas

GenAI

Online Learning

Deep RL

Meta-Learning

NLP

Autonomous Navigation

Applied ML

ML/DL Frameworks

PyTorch

TensorFlow

Hugging Face Transformers

BERT family

FAISS

Cloud & MLOps

Microsoft Azure

Docker

Languages

Python

SQL

C

JavaScript

MATLAB

LANGUAGES

Arabic – Native English – Fluent